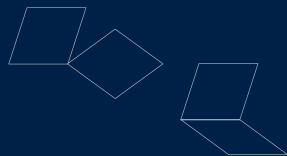


Random Walks on Cayley Graphs

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Motivation - Random Number Generators

- ▶ The need to generate random numbers: Cryptography, Statistical methods, etc.
- ▶ Any algorithm by which a computer generates a random number must be deterministic. Every such algorithm needs a seed, with which it starts, and by knowing the seed, the results can be replicated.
- ▶ The only way to get a truly random number from an algorithm is to input a random seed into the algorithm.
- ▶ In 1955, the RAND Corporation published a table of a million random digits, which was widely used for the same.

Motivation - Random Number Generators

- The random seed can be taken from physical processes as well. Cloudflare - a cybersecurity company - has a wall of lava lamps - they take photos of the lamps and extract random data from that - to seed random number generating algorithms - to get keys for encryption.



Source: <https://www.cloudflare.com/learning/ssl/lava-lamp-encryption/>



Motivation - Random Number Generators

- ▶ We can still get pretty close to random by various mathematical models, one of which we will see today!!
- ▶ Most Random Number Generators are special cases of the general scheme:
Start with a modulus m , a multiplier a , an increment b , and an initial value X_0 .
Create a sequence X_n , $n = 0, 1, 2, 3, \dots$ recursively as follows:

$$X_{n+1} = aX_n + b \mod m$$

- ▶ $m = 2^{32}$ or $m = 2^{31} - 1$ are popular choices.
- ▶ D.H. Lehmer introduced this with $b = 0$.



What is a Markov Chain?

Consider a sequence of random variables $X_1, X_2, \dots$ taking values in the state space, $\{s_1, s_2, \dots, s_m\}$. If $P(X_{n+1} = s_j \mid X_n = s_i) = t_{ij}$ for all $n \in \mathbb{N}$, $i, j \in \{1, 2, \dots, m\}$, then this sequence is called a **Markov Chain**. The matrix, $T = [t_{ij}]_{m \times m}$ is called the corresponding **Transition matrix**.

Here, n represents the step we are in, and X_n is the state observed in that step. If we consider the probability vectors, $P_k = [p_1, p_2, \dots, p_m]$, where p_i represents the probability of being in the state s_i in step k , we can find the probability vector of the general step P_n as follows:

$$P_n = TP_{n-1} \implies P_n = T^n P_0$$

How?

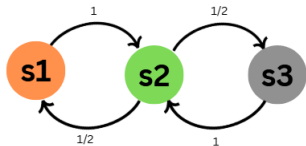
Given that P_0 is the initial probability vector (in state 0), the above equations hold for all $n \in \mathbb{N}$.



The transition graph of a Markov Chain

Consider a weighted directed graph. The vertices can be the states and the weights of the edges can give us the probabilities of each transition.

For example, consider the following graph:



We get the transition matrix: $T = \begin{pmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{pmatrix}$

If the process starts at s_i , then $P_0 = e_i$, for $i \in \{1, 2, 3\}$, and we can find P_n for any $n \in \mathbb{N}$ using $P_n = TP_{n-1} \implies P_n = T^n e_i$.

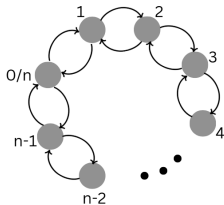


Cayley Graphs

What is a Cayley Graph?

Let H be a finite group, and S a subset of it. The corresponding Cayley Graph, $G = \chi(H, S)$ is the graph in which $V(G) = H$ and two vertices $g, h \in H$ are joined by a directed edge from g to h iff $\exists s \in S$ such that $g = sh$.

For example: $\chi(\mathbb{Z}/n\mathbb{Z}, S)$, where $S = \{\pm 1 \pmod n\}$ is the cyclic graph on n vertices:



Even though Cayley Graphs are generally directed, in this lecture, we will be looking at undirected ones. (S : symmetric)



Random Walks on the Finite Circle

Consider a creature moving on the graph $\chi(\mathbb{Z}/n\mathbb{Z}, S)$, $S = \{\pm 1 \bmod n\}$, going from vertex x to either $x - 1$ or $x + 1$, in each step, with equal probability. This gives rise to a Markov Chain with the following properties:

$$\blacktriangleright T = \begin{pmatrix} 0 & 1/2 & 0 & 0 & \dots & 0 & 1/2 \\ 1/2 & 0 & 1/2 & 0 & \dots & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 & \dots & 0 & 0 \\ 0 & 0 & 1/2 & 0 & \dots & 0 & 0 \\ \cdot & \cdot & \cdot & \cdot & \dots & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \dots & \cdot & \cdot \\ 1/2 & 0 & 0 & 0 & \dots & 1/2 & 0 \end{pmatrix}$$

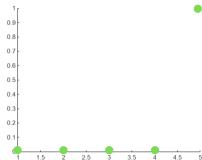
- $\blacktriangleright$ If A is the adjacency matrix of the graph, and T is the Markov transition matrix, then $T = \frac{1}{2}A$

This is an example of a random walk on the finite circle.

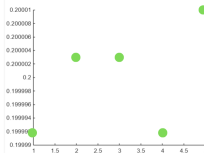


Random Walks on the Finite Circle

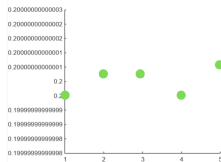
Taking the random walk on the finite circle with 5 vertices, if the creature starts walking from 0 or 5, we get the following probabilities:



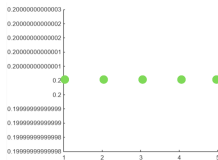
t=0



t=50



t=150



t=200

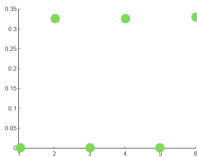


Random Walks on the Finite Circle

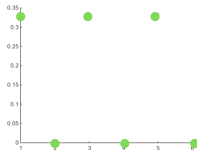
Taking the random walk on the finite circle with 6 vertices, if the creature starts walking from 0 or 6, we get the following probabilities:



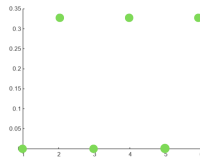
t=0



t=50



t=99



t=100



Random Walks on the Finite Circle

In general, consider the graph, $G = \chi(\mathbb{Z}/n\mathbb{Z}, S)$, with the adjacency matrix, A . Let k be the degree of the graph, i.e. $k = |S|$.

We get a Markov Chain, with the states as the set of vertices, and the probability in any given step, of going to a connected vertex as $\frac{1}{k}$.

Using the same notation as before, we get $T = \frac{1}{k}A$.

Additionally, let $p^t(x)$ be the probability density on $x \in X$, at time t . Then $p^t(x) \geq 0$ for all $x \in X, t \geq 0$, and $\sum_{x \in X} p^t(x) = 1$.

Here $p^t(x)$ is the probability that our random walker is at state x at step/time t . Note that we consider everything to be discrete here.



Spectral Theorem and some related results

The **Spectral Theorem** states that:

Given a real symmetric matrix, A of dimension $(n \times n)$, there exists an orthonormal basis of $\mathbb{R}^n$, which is made up of the eigenvectors of this matrix, A . We represent this basis as $\{\phi_1, \phi_2, \dots, \phi_n\}$. Let X be a k -regular graph with the adjacency operator, A . Here we are interested in looking at the spectrum of the graph.

- ▶ The degree k is an eigenvalue of A , and $|\lambda| \leq k$ for all eigenvalues of A .
- ▶ The k -regular graph, X is bipartite iff $-k$ is an eigenvalue of the adjacency operator.
- ▶ The graph is connected iff k has multiplicity 1 as an eigenvalue of A .

If T is the Markov Transition matrix, then $T = \frac{1}{k}A$
 $\implies \text{spectrum}(T) = \frac{1}{k} \text{spectrum}(A)$.



Random Walks Approach the Uniform Distribution

Theorem 1

Suppose X is a connected, nonbipartite regular graph of degree k with n vertices. Let A be the adjacency operator of X and $T = \frac{1}{k}A$, the Markov transition operator. Then for every probability function on X , that is, $p(x) \geq 0$, for all $x \in X$, and $\sum_{x \in X} p(x) = 1$, we have

$$\lim_{t \rightarrow \infty} T^t p = u$$

where the uniform distribution is

$$u(x) = \frac{1}{n}, \text{ for all } x \in X.$$



Random Walks Approach the Uniform Distribution

Proof.

Spectral Theorem for $T \implies \exists$ an orthonormal basis $\{\phi_1, \phi_2, \dots, \phi_n\}$ in $L^2(X)$ s.t.

1. $T\phi_j = \lambda_j\phi_j$, where $j = 1, 2, \dots, n$
2. $\langle \phi_i, \phi_j \rangle = \delta_{ij}$, where $i, j \in \{1, 2, \dots, n\}$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are eigenvalues of T . Also for any $f \in L^2(X)$,

$$f(x) = \sum_{i=1}^n \langle f, \phi_i \rangle \phi_i(x)$$

Hence, for any p , which is a probability function, $p(x) = \sum_{i=1}^n \langle p, \phi_i \rangle \phi_i(x)$

$$\implies Tp(x) = \sum_{i=1}^n \langle p, \phi_i \rangle \lambda_i \phi_i(x)$$

$$\implies T^t p(x) = \sum_{i=1}^n \langle p, \phi_i \rangle \lambda_i^t \phi_i(x)$$



Random Walks Approach the Uniform Distribution

Proof.

Since the graph is connected and nonbipartite,

$$\lambda_1 = 1 > \lambda_2 \geq \dots \geq \lambda_n > -1$$

How?

Hence, we get

$$\lim_{t \rightarrow \infty} \lambda_i^t = \delta_{1i}$$

which gives us the final result we need:

$$\lim_{t \rightarrow \infty} T^t p = \lim_{t \rightarrow \infty} \sum_{i=1}^n \langle p, \phi_i \rangle \lambda_i^t \phi_i = \langle p, \phi_1 \rangle \phi_1 = \langle p, \frac{1}{\sqrt{n}} \rangle \frac{1}{\sqrt{n}} = \langle p, 1 \rangle \frac{1}{n} = \frac{1}{n} = u$$





Comparison of norms on $\mathbb{R}^n$

Let X be our graph. We define the L^1 -norm, inner product and L^2 -norm of real-valued functions on X , f, g as follows:

- ▶ $\|f\|_1 = \sum_{x \in X} |f(x)|$
- ▶ $\langle f, g \rangle = \sum_{x \in X} f(x)g(x)$
- ▶ $\|f\|_2 = \langle f, f \rangle^{1/2}$

Proposition 1

$$\|f\|_2 \leq |X|^{1/2} \|f\|_1$$

Proof.

Taking $g(x) = 1$ for all $x \in X$, in the Cauchy-Schwarz inequality, $|\langle f, g \rangle| \leq \|f\|_2 \|g\|_2$, we get the required result. $\square$



Random Walks Approach the Uniform Distribution

After how many steps is the probability distribution of the creature close to random?

Theorem 2

Suppose we have a connected nonbipartite k -regular graph, X with n vertices. Let the Markov transition operator on X be $T = \frac{1}{k}A$, where A is the adjacency operator.

Define $\beta = \max\{|\lambda| \mid \lambda \in \text{spectrum}(T), \lambda \neq 1\}$.

Then for every probability function, p on X , that is $p(x) \geq 0$, for all $x \in X$, and $\sum_{x \in X} p(x) = 1$, we have

$$\|T^m p - u\|_1 \leq \sqrt{n} \beta^m$$

where u is the uniform probability distribution.

Proof.

Let $\{\phi_1, \phi_2, \dots, \phi_n\}$ be the orthonormal basis of T , used in proof of Theorem 1. Applying the proposition, relating the L^1 and L^2 norms, we get

$$\|T^m p - u\|_1^2 \leq n \|T^m p - u\|_2^2$$

$$\text{Now, } \|T^m p - u\|_2^2 = \|\sum_{j=2}^n \langle p, \phi_j \rangle \lambda_j^m \phi_j\|_2^2 = \sum_{j=2}^n |\langle p, \phi_j \rangle|^2 |\lambda_j|^{2m}$$

$$\implies \|T^m p - u\|_1^2 \leq n \beta^{2m} \sum_{j=1}^n |\langle p, \phi_j \rangle|^2 = n \beta^{2m} \|p\|_2^2$$

$$\text{Also, } \|p\|_2^2 = \sum_{x \in X} |p(x)|^2 \leq \sum_{x \in X} |p(x)| = 1.$$

$$\text{Hence, } \|T^m p - u\|_1^2 \leq n \beta^{2m}$$

$$\implies \|T^m p - u\|_1 \leq \sqrt{n} \beta^m$$





- ▶ How can we generate random numbers? - A general scheme
- ▶ Some background - Markov Chains, Cayley Graphs, and Spectral Theory
- ▶ Main result: For connected nonbipartite, regular graphs, the probability vector of a random walk approaches the uniform distribution
- ▶ Rate of this convergence



- ▶ Fourier Analysis on Finite Groups and Applications, Audrey Terras.